

$$m \frac{d^2 \vec{r}}{dt^2} = q \vec{v} \times \vec{B}$$

$$= B_0 q \vec{v} \times \hat{e}_z$$

$$\vec{v}_0 \cdot \hat{e}_z = 0$$

• $\vec{v}_0$ is orthogonal to $\hat{e}_z$

$$\vec{v} \times \vec{e}_z = \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ v_x & v_y & v_z \\ 0 & 0 & 1 \end{vmatrix} = v_y \hat{e}_x - v_x \hat{e}_y$$

$$\vec{r} = (x(t), y(t), z(t))$$

$$\frac{d\vec{r}}{dt} = (v_x(t), v_y(t), v_z(t))$$

$$\frac{d\vec{v}}{dt} = \left(\frac{dv_x}{dt}, \frac{dv_y}{dt}, \frac{dv_z}{dt} \right)$$

$$\frac{d\vec{v}}{dt} = \left(\frac{B_0 q}{m} \right) (v_y, -v_x, 0)$$

